Supplementary: subdistribution HR (Fine–Gray) by progression pattern (reference: IH-NL)

Fine–Gray via survival::finegray with stratum-specific censoring weights; robust SE

	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	1.08 (0.88–1.34)	1.50 (0.78–2.89)	1.16 (0.75–1.81)	0.98 (0.51–1.89)
IH-NL	Reference	Reference	Reference	Reference
IH-TG+NL	1.49 (1.08–2.07)*	1.82 (0.71–4.66)	1.12 (0.59–2.13)	0.80 (0.27–2.37)
EH-TG	1.10 (0.76–1.59)	3.33 (1.59–6.96)**	1.39 (0.78–2.46)	0.83 (0.26–2.66)
EH-NL	0.71 (0.54–0.93)*	2.78 (1.51–5.14)**	1.91 (1.25–2.92)**	1.62 (0.87–3.02)
EH-TG+NL	0.82 (0.42–1.59)	2.59 (0.74–9.09)	1.94 (0.84–4.48)	3.20 (0.89–11.52)
Sys-TG	1.20 (0.79–1.82)	1.40 (0.44–4.51)	2.81 (1.55–5.07)***	NE (1 event)
Sys-NL	0.74 (0.49–1.10)	4.35 (2.01–9.40)***	2.68 (1.58–4.55)***	NE (2 events)
Sys-TG+NL	0.93 (0.69–1.26)	2.14 (1.05–4.36)*	2.80 (1.77–4.44)***	0.85 (0.33–2.20)
Overall Wald P (df=8)	**P = 0.003**	**P = 0.003**	**P < 0.001**	**P = 0.256**

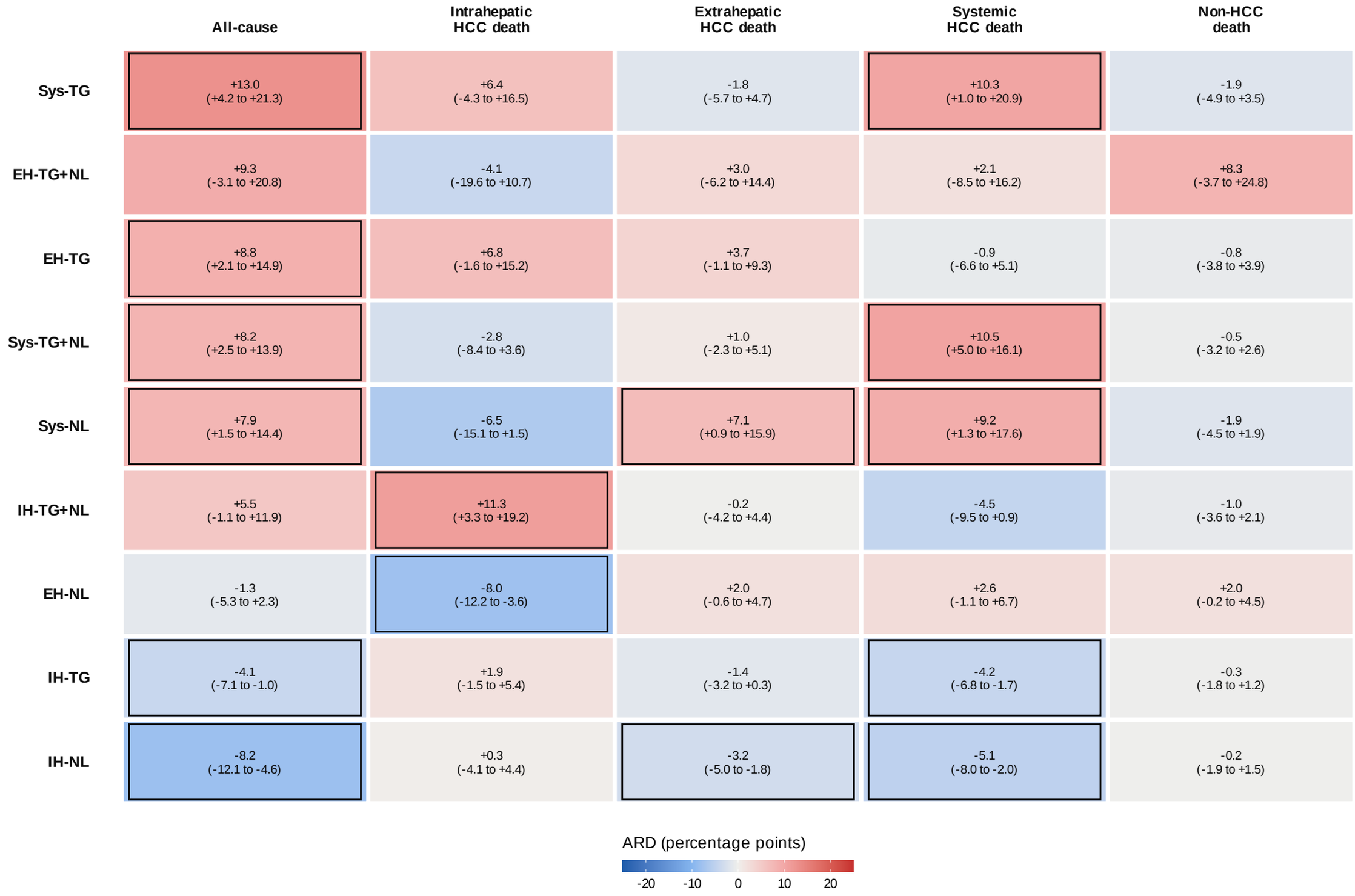
$\log_2(\text{sHR})$

-2 -1 0 1 2

Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: absolute risk difference at 12 months vs population-standardised average

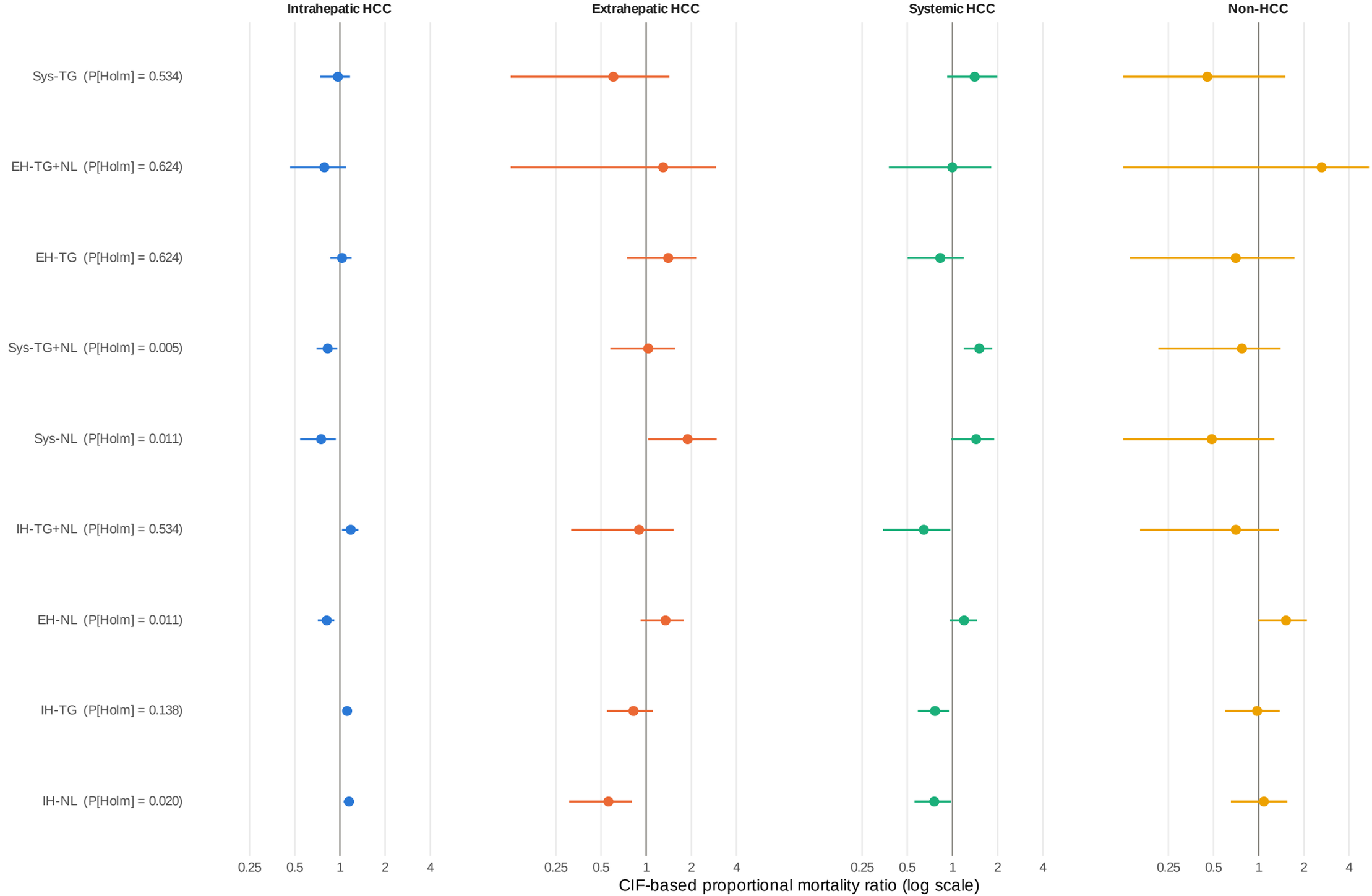
ARD = standardised CIF of the pattern − weighted average over patterns (weights = observed pattern shares); boxed: 95% CI excludes 0



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: cause-of-death enrichment (PMR) at 12 months

PMR = share of deaths from cause j within the pattern / share in all progressors. 95% bootstrap CI (clipped at 1/8 and 8).
P[Holm]: weighted-effect-coded Wald test (df = 2) that the pattern's csHR is equal across IH/EH/Systemic HCC death (from Lunn–McNeil model).

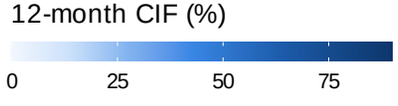


PMR is a proportion metric, not a risk metric: interpret alongside the absolute CIF matrix; PMRs within a pattern are mutually constrained.

Supplementary: unadjusted 12-month cumulative incidence (Aalen–Johansen)

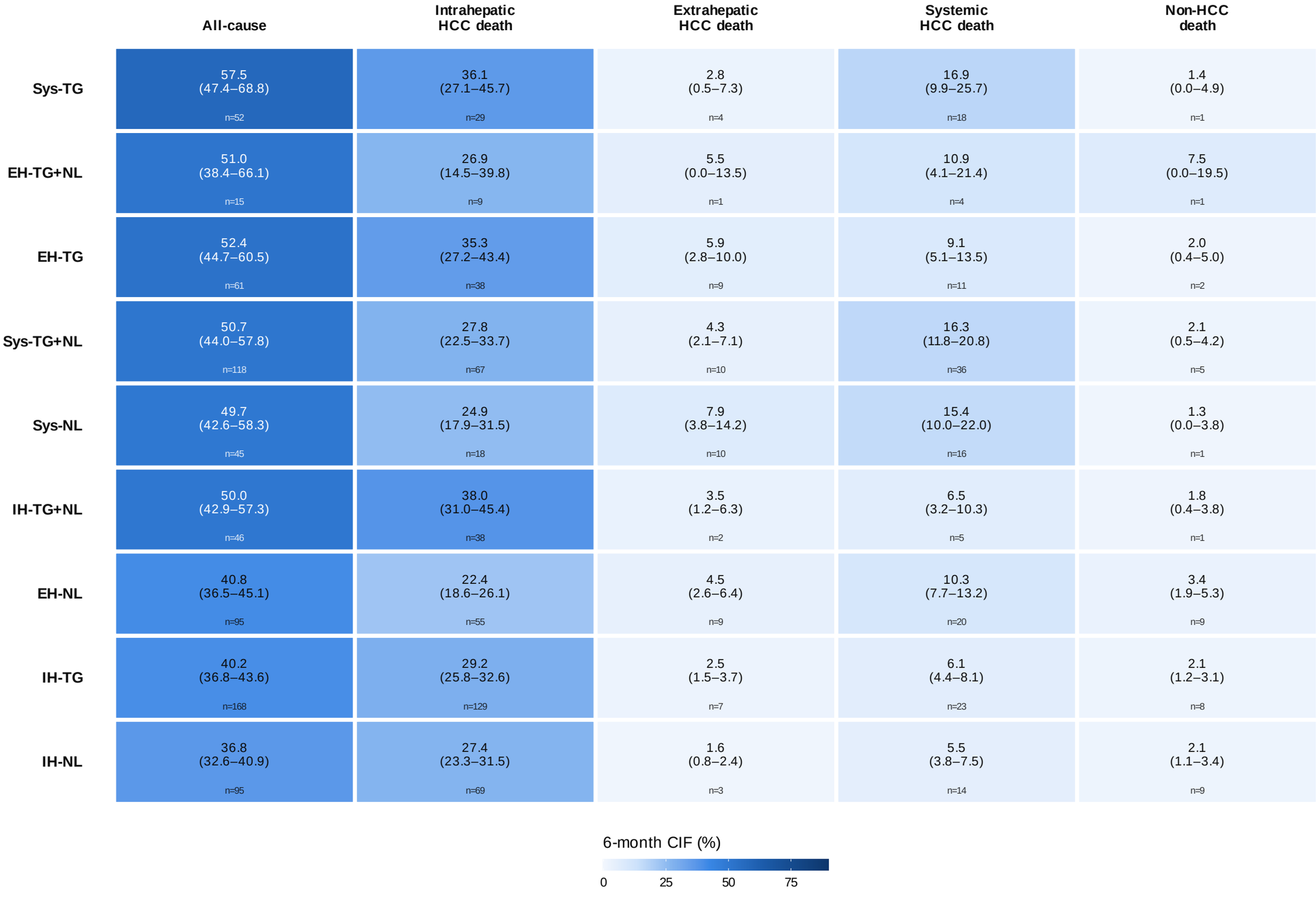
% (95% CI); all-cause = 1 − Kaplan–Meier; n = events within 12 months

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
Sys-TG	83.4 (70.5–90.7) n=55	47.5 (36.6–61.7) n=31	5.7 (2.2–14.8) n=4	28.6 (19.5–41.8) n=19	1.6 (0.2–11.2) n=1
EH-TG+NL	74.0 (51.2–86.2) n=21	35.0 (21.2–57.8) n=10	7.2 (1.9–27.3) n=2	20.9 (10.3–42.8) n=6	10.9 (3.7–31.7) n=3
EH-TG	63.4 (53.1–71.4) n=72	36.4 (28.6–46.4) n=42	10.0 (5.7–17.5) n=11	13.5 (8.4–21.6) n=15	2.7 (0.9–8.1) n=3
Sys-TG+NL	78.1 (70.7–83.7) n=135	43.3 (36.4–51.4) n=75	7.0 (4.1–12.1) n=12	24.3 (18.7–31.7) n=42	3.5 (1.6–7.7) n=6
Sys-NL	81.0 (69.2–88.2) n=63	33.0 (24.1–45.4) n=26	16.8 (10.2–27.7) n=13	28.5 (19.9–40.6) n=22	2.7 (0.7–10.6) n=2
IH-TG+NL	73.0 (62.3–80.7) n=74	53.8 (44.7–64.6) n=55	6.2 (2.9–13.6) n=6	9.9 (5.5–17.8) n=10	3.2 (1.0–9.7) n=3
EH-NL	62.9 (56.2–68.6) n=159	29.3 (24.2–35.5) n=76	7.3 (4.7–11.4) n=18	19.7 (15.3–25.5) n=48	5.9 (3.6–9.6) n=15
IH-TG	66.8 (61.7–71.3) n=269	47.4 (42.7–52.6) n=193	5.0 (3.2–7.7) n=19	9.6 (7.1–13.1) n=38	4.0 (2.5–6.5) n=16
IH-NL	49.8 (43.9–55.1) n=161	33.4 (28.6–39.0) n=109	2.6 (1.3–5.1) n=8	9.2 (6.5–13.0) n=29	4.7 (2.8–7.7) n=15



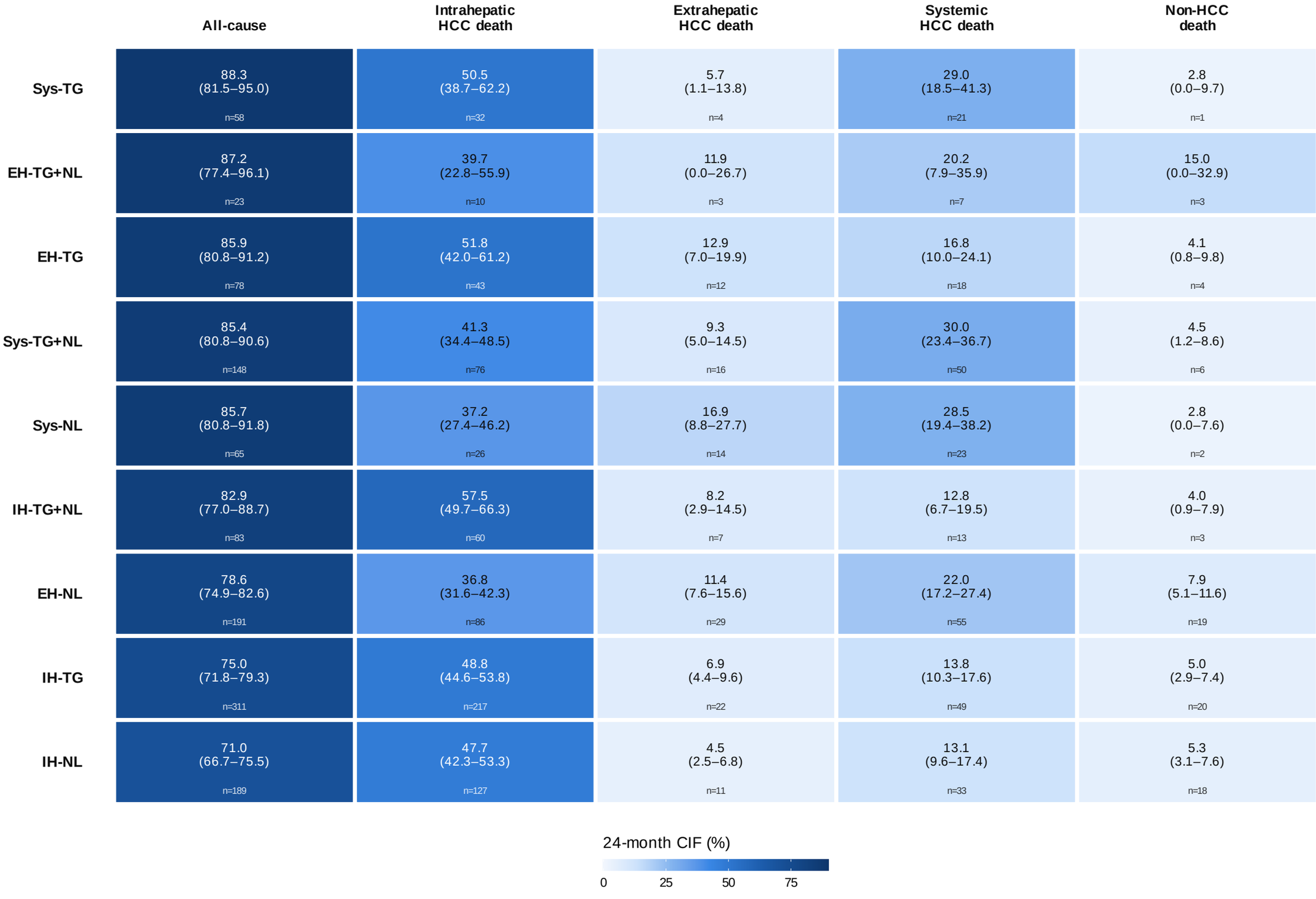
Supplementary: adjusted 6-month CIF after first progression

Same models as Figure 4.4A; n = events within 6 months



Supplementary: adjusted 24-month CIF after first progression

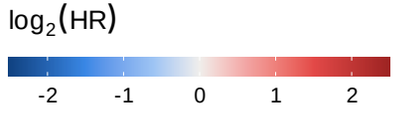
Same models as Figure 4.4A; n = events within 24 months



Supplementary: Firth-penalised HR / csHR (reference: IH-NL)

coxphf (Firth-penalised partial likelihood, Wald CI from penalised information); treatment group entered as covariate (coxphf has no strata)

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	1.17 (0.98–1.40)	1.10 (0.89–1.37)	1.68 (0.87–3.26)	1.18 (0.75–1.84)	1.01 (0.52–1.95)
IH-NL	Reference	Reference	Reference	Reference	Reference
IH-TG+NL	1.55 (1.20–2.00)***	1.59 (1.17–2.16)**	2.58 (1.05–6.34)*	1.46 (0.76–2.77)	1.07 (0.37–3.13)
EH-TG	1.81 (1.39–2.35)***	1.55 (1.09–2.20)*	4.48 (2.14–9.36)***	2.06 (1.16–3.66)*	1.19 (0.40–3.55)
EH-NL	1.26 (1.04–1.54)*	0.84 (0.64–1.11)	3.06 (1.63–5.73)***	1.96 (1.27–3.01)**	1.64 (0.85–3.16)
EH-TG+NL	1.78 (1.15–2.76)*	1.24 (0.65–2.36)	4.84 (1.42–16.56)*	2.38 (1.04–5.47)*	5.04 (1.48–17.22)**
Sys-TG	2.22 (1.64–2.99)***	1.82 (1.23–2.70)**	2.49 (0.82–7.54)	4.10 (2.33–7.21)***	NE (1 event)
Sys-NL	1.75 (1.32–2.32)***	1.08 (0.71–1.63)	6.18 (2.90–13.18)***	3.35 (1.95–5.74)***	NE (2 events)
Sys-TG+NL	1.73 (1.40–2.15)***	1.22 (0.91–1.62)	3.09 (1.50–6.41)**	3.52 (2.25–5.49)***	1.25 (0.48–3.23)
Overall Wald P (df=8)	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.310**



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: HR / csHR excluding undetermined deaths

n = 1778 (8 deaths with unclassified/unknown/missing cause removed)

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	1.16 (0.97–1.39)	1.10 (0.89–1.38)	1.68 (0.87–3.25)	1.18 (0.75–1.84)	1.02 (0.53–1.96)
IH-NL	Reference	Reference	Reference	Reference	Reference
IH-TG+NL	1.58 (1.22–2.04)***	1.64 (1.21–2.23)**	2.49 (1.00–6.22)	1.39 (0.73–2.67)	0.99 (0.33–2.97)
EH-TG	1.80 (1.38–2.35)***	1.56 (1.10–2.22)*	4.42 (2.11–9.25)***	2.00 (1.11–3.58)*	1.07 (0.35–3.28)
EH-NL	1.24 (1.02–1.52)*	0.83 (0.63–1.09)	2.97 (1.58–5.58)***	1.94 (1.26–3.00)**	1.66 (0.87–3.18)
EH-TG+NL	1.71 (1.10–2.66)*	1.13 (0.58–2.17)	4.03 (1.10–14.71)*	2.30 (0.98–5.41)	4.03 (1.12–14.48)*
Sys-TG	2.17 (1.61–2.93)***	1.74 (1.17–2.58)**	2.19 (0.70–6.84)	4.14 (2.35–7.30)***	NE (1 event)
Sys-NL	1.70 (1.28–2.25)***	1.02 (0.68–1.55)	5.78 (2.69–12.41)***	3.28 (1.91–5.65)***	NE (2 events)
Sys-TG+NL	1.70 (1.37–2.12)***	1.18 (0.88–1.57)	3.14 (1.51–6.50)**	3.55 (2.27–5.55)***	1.12 (0.43–2.92)
Overall Wald P (df=8)	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.411**



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: HR / csHR including patients flagged for deletion

n = 1797

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	1.14 (0.96–1.37)	1.09 (0.87–1.35)	1.68 (0.87–3.26)	1.16 (0.74–1.81)	1.01 (0.53–1.93)
IH-NL	Reference	Reference	Reference	Reference	Reference
IH-TG+NL	1.57 (1.21–2.02)***	1.65 (1.22–2.25)**	2.54 (1.02–6.34)*	1.43 (0.75–2.74)	1.01 (0.34–3.01)
EH-TG	1.79 (1.38–2.34)***	1.58 (1.11–2.24)*	4.52 (2.16–9.46)***	2.04 (1.14–3.65)*	1.10 (0.36–3.36)
EH-NL	1.25 (1.03–1.52)*	0.84 (0.64–1.10)	2.98 (1.59–5.60)***	1.94 (1.26–3.00)**	1.68 (0.88–3.20)
EH-TG+NL	1.69 (1.08–2.63)*	1.14 (0.59–2.20)	4.15 (1.14–15.12)*	2.37 (1.01–5.57)*	4.14 (1.15–14.88)*
Sys-TG	2.15 (1.59–2.90)***	1.75 (1.18–2.60)**	2.24 (0.72–6.98)	4.19 (2.38–7.38)***	NE (1 event)
Sys-NL	1.68 (1.27–2.22)***	1.03 (0.68–1.56)	5.89 (2.74–12.64)***	3.32 (1.93–5.71)***	NE (2 events)
Sys-TG+NL	1.67 (1.34–2.07)***	1.16 (0.87–1.54)	3.15 (1.52–6.53)**	3.49 (2.23–5.46)***	1.13 (0.44–2.94)
Overall Wald P (df=8)	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.385**



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.